WORLD RESOURCES INSTITUTE

TECHNICAL NOTE

# Climate Watch NDC Tracker

## Estimating global and national GHG emissions based on emissions reduction targets in countries' Nationally Determined Contributions

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*Technical notes document the research or analytical methodology underpinning a publication, interactive application, or tool.*

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## Abstract

Understanding future greenhouse gas (GHG) emissions trajectories is crucial for addressing global climate change. Climate Watch is an online data platform that tracks global and country GHG emissions, as well as climate commitments. This technical note describes a methodology used by Climate Watch for estimating future global and country-level GHG emissions based on GHG targets communicated in countries' Nationally Determined Contributions (NDCs). It aims to estimate countries' economy-wide GHG emissions for target years, assuming NDC targets are fully achieved. For sectors and GHGs not covered by NDCs, supplemental data sources are used. Global GHG emissions are aggregated from country-level estimates, accounting for emissions not covered by NDCs, such as international bunkers. This analysis supports the understanding of NDCs, the tracking of progress towards national and global climate goals, and helps to identify gaps in global emissions pathways. The methodology underpins data and visualizations made available through Climate Watch's NDC Tracker.

## Introduction

The Paris Agreement—a legally binding international treaty on climate change adopted by 196 Parties at COP21 in December 2015—establishes the overarching goal of keeping the global average temperature rise to well below 2°C above pre-industrial levels, while striving to limit the increase to 1.5°C (UNFCCC 2015). Since the adoption of the Paris Agreement, countries have been submitting national climate action plans, known as Nationally Determined Contributions (NDCs), that outline their pledges to reduce greenhouse gas (GHG) emissions and strengthen resilience against the impacts of climate change.

Under the Paris Agreement, Parties are required to communicate and update their NDCs every five years, with each submission representing greater ambition than the previous one (UNFCCC 2015). In line with this requirement,

most Parties submitted their initial commitments in 2015 and have continued to update them in successive cycles. The newer rounds of NDCs introduce new GHG targets for future years and may include revisions to existing targets.

Recent scientific findings indicate that to limit warming to 1.5°C,[1] global GHG emissions must peak by 2025, declining by 42 percent by 2030[2] and by 57 percent by 2035, compared to 2019 levels (UNEP 2024). Delaying or failing to achieve those milestones would threaten the achievement of the goals of the Paris Agreement and leave the 1.5°C target out of reach. To assess whether GHG emissions are on track, understanding the emissions impacts of NDCs is essential. However, there are challenges to estimating GHG emissions under NDCs, including diverse mitigation commitments and incomplete target coverage. GHG targets can also be framed in different ways, posing different calculation and data requirements. Additionally, NDCs might not cover all sectors and GHGs, or may lack specification of GHG target coverage. Researchers must therefore make assumptions to fill in missing emissions and complete economy-wide estimates.

Climate Watch is an online data platform that offers data, tools, and visualizations to track global climate progress. It highlights GHG emissions trends and provides climate-relevant data such as emissions drivers and climate vulnerabilities. Additionally, it tracks countries' climate commitments, including NDCs and long-term low-emission development strategies. A key feature of Climate Watch is the tracking of NDCs, providing structured indicators that facilitate the understanding of NDCs and their targets. While many NDCs include GHG targets, they might also contain other mitigation commitments that are not framed in terms of GHG emissions reductions—also referred to as non-GHG targets—such as the scale-up of renewables or reduction of deforestation. Furthermore, NDCs often contain adaptation components addressing resilience, vulnerabilities, and climate risks. The methodology discussed here, however, focuses exclusively on estimating emissions levels based on GHG targets in NDCs.

This technical note describes the methodology of the Climate Watch NDC Tracker's estimation of global and country-level GHG emissions, based on targets communicated in countries' NDCs. When calculating emissions, it is assumed that GHG targets in NDCs are achieved in full in target years. This analysis supports assessment of the achievement of countries' NDCs, facilitates comparison of GHG emissions under the NDCs, and helps to identify gaps between global GHG emissions trajectories and that which is needed to achieve global temperature goals. By publishing this technical note, we aim to provide transparency regarding our dataset, to complement other research efforts (den Elzen et al. 2022; Climate Analytics and NewClimate Institute 2024; IEA 2024), to contribute to a broader understanding of emissions estimates, and to facilitate comparisons with other methodologies.

The aim of the methodology is to estimate economy-wide GHG emissions for each country for target years (2030, 2035, and so on). Additionally, the methodology produces a global emissions estimate, aggregating country-level emissions and accounting for emissions not covered by NDCs. The following steps are taken to estimate country-level (Steps 1 and 2) and global (Step 3) GHG emissions for target years associated with NDCs:

1. Analyzing GHG targets in NDCs
    i. Determine if the NDC contains GHG targets and identify target types
    ii. Determine the sector and gas coverage of GHG targets
2. Estimating country-level GHG emissions under NDCs
    i. Calculate target-year emissions for the sectors and GHGs covered by the target, based on GHG target types, using data from NDCs and emissions datasets (e.g., national inventory reports, Climate Watch)
    ii. Quantify target-year emissions for the sectors and GHGs not covered by the target or for missing targets, using emissions projections derived from the Network for Greening the Financial System (NGFS) Climate Scenarios data (Richters et al. 2024)
    iii. Produce economy-wide GHG emissions calculations under NDCs for all countries
3. Estimating global GHG emissions under NDCs
    i. Aggregate country-level emissions to produce a global total
    ii. Adjust global emissions for international bunker fuels that are not covered in any NDC, assuming achievement of International Civil Aviation Organization (ICAO) and International Maritime Organization (IMO) emissions targets
    iii. Adjust global emissions for the differences in land use, land-use change and forestry (LULUCF) emissions between national inventories and bookkeeping models

The details of these steps are described in the sections which follow.

# Analyzing GHG targets in NDCs

All NDCs contain mitigation commitments to reduce GHG emissions (Climate Watch 2025). Those commitments can take the form of quantified numerical targets (GHG or non-GHG targets), or sectoral policies and actions (Figure 1). **GHG targets** refer to contributions framed as numerical outcomes in terms of GHGs (including base-year targets, fixed-level targets, baseline scenario targets, intensity targets, and trajectory targets). **Non-GHG targets** refer to contributions that are framed in terms of quantifiable outcomes in areas such as energy efficiency, renewable energy, forestry, or others that are not expressed in terms of emissions or emissions reductions. **Policies and actions** refer to intentions or commitments to implement those measures that will help achieve GHG emission reductions (Fransen et al. 2021). This technical note focuses on the estimation of countries' GHG emissions levels based on targets communicated in their NDCs.

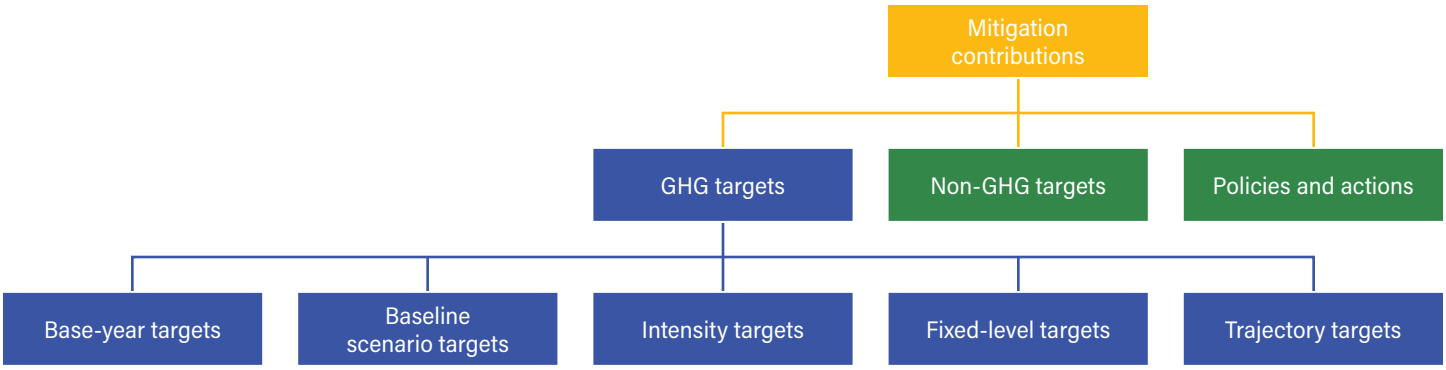
We do not calculate the emissions impact of non-GHG targets and sectoral policies and actions. When non-GHG targets and sectoral measures overlap in scope with GHG targets, it is assumed that the impact of those measures is already accounted for within the target. For example, if a country targets a certain amount of solar energy capacity, we assume that the GHG benefits of achieving the target is counted within that country's target for overall emissions reductions in the energy sector. If non-GHG targets and measures fall outside the scope of the GHG target, we do not estimate their impact in terms of GHG emissions, as this exceeds the scope of our methodology. For example, if a country has a target to reforest a certain number of hectares, but their GHG target does not cover LULUCF, we do not attempt to quantify the emissions impact of the reforestation efforts. This results in a conservative emissions estimate at country and global levels.

## Types of GHG targets

GHG targets can take various forms (Figure 1), each with specific data requirements and calculation methods. The target types and definitions (GHG Protocol 2014; Levin et al. 2015; Fransen et al. 2021) are as follows:

- **Base-year target:** A commitment to reduce—or control the increase of—emissions by a specified quantity relative to a base year. Example: United Arab Emirates commits to reduce its emissions by 19 percent by 2030, relative to 2019 levels.
- **Baseline scenario target:** A commitment to reduce emissions by a specified quantity relative to a projected baseline scenario. Example: Guatemala commits to reducing its emissions by 11.2 percent (unconditional) and by 22.6 percent (conditional) by 2030, compared to business as usual (BAU).
- **Intensity target:** A commitment to reduce emissions intensity (emissions per unit of another variable, typically gross domestic product [GDP] or population) by a specified quantity. Example: India commits to reducing its emissions intensity per unit of GDP by 45 percent by 2030, compared to 2005 levels.
- **Fixed-level target:** A commitment to reduce—or control the increase of—emissions to an absolute emissions level in a target year or target period. A carbon budget, for example, is a multi-year fixed-level target. Example: Argentina commits to an unconditional, economy-wide cap on net emissions of 349 million tons of carbon dioxide equivalent ($MtCO_2e$) in 2030.

Figure 1 | **Types of mitigation contribution**




*Source:* Fransen et al. (2021), adapted from Levin et al. (2015).

- **Trajectory target:** A commitment to reduce—or control the increase of—emissions in multiple target years or over a period of time longer than a year (this includes peaking targets). Example: In 2025, South Africa's annual GHG emissions will range from 398–510 $MtCO_2e$. In 2030, South Africa's annual GHG emissions will range from 350–420 $MtCO_2e$.

A handful of NDCs do not contain GHG targets but provide explicit information regarding the emissions impacts of the measures contained in the NDCs. For those NDCs, we attempt to treat the impact as a GHG target, and apply the same methodology detailed below for estimating economy-wide GHG emissions levels. For example: Benin commits to measures which are estimated to reduce its emissions by 20.15 percent by 2030, compared to BAU. This is treated as a baseline scenario target in our calculations.

## Sector and gas coverage of GHG targets

This technical note aims to produce comprehensive economy-wide GHG emissions estimates for countries, based on GHG targets communicated in their respective NDCs. A GHG target is economy-wide if it covers all sectors and gases. We therefore examine the sector and gas coverage—also referred to as a target boundary—of GHG targets in NDCs and make assumptions for sectors and gases that are not covered. For sectors and gases outside of target boundaries, emissions projections adapted from the NGFS Current Policies scenario (Richters et al. 2024) are applied.

NDCs typically specify the sector and gas coverage of the GHG targets to ensure clarity, transparency, and understanding. Sectors include all potential sources and sinks of GHG emissions, including energy, industrial processes, waste, agriculture, and land-use, land-use change and forestry (LULUCF). Gases considered include carbon dioxide ($CO_2$), methane ($CH_4$), nitrous oxide ($N_2O$), hydrofluorocarbons (HFCs), perfluorocarbons (PFCs), sulfur hexafluoride ($SF_6$), and nitrogen trifluoride ($NF_3$). HFCs, PFCs, $SF_6$, and $NF_3$ are often collectively referred to as fluorinated gases, or F-gases.

To assess sector coverage, we follow Intergovernmental Panel on Climate Change (IPCC) guidelines for emissions reporting to the extent possible. This ensures consistency and comprehensiveness when combining data from different historical data sources and projections. In cases where an NDC does not fully adhere to IPCC categorizations or only covers part of a sector, we apply approximations as applicable. For example, if the target includes two energy subsectors (electricity and transport), we examine the historical emissions inventory to determine if these two subsectors represent the majority of energy emissions. If the majority of energy sector emissions are covered, we consider the target as covering the entire energy sector. If not, we fill in gaps for other energy subsectors, as allowed by data availability.

While economy-wide targets should cover all seven GHGs—$CO_2$, $CH_4$, $N_2O$, HFCs, PFCs, $SF_6$, $NF_3$—we consider a target as covering all gases as long as it covers $CO_2$, $CH_4$, $N_2O$, and HFCs. F-gases constitute less than 4 percent of total GHG emissions, and HFCs constitute more than 90 percent of F-gases (Climate Watch 2024). The effect of this assumption on our estimates is thus negligible.

In addition, when an NDC is unclear about the sector and gas coverage of its GHG target, we assume that the target is economy-wide, covering all sectors and gases. The Paris Agreement and its subsequent decisions, including those made during the Global Stocktake, emphasize the need for an economy-wide target (UNFCCC 2024b). Article 4 of the Paris Agreement requires developed countries to establish economy-wide emissions reduction targets, while encouraging developing countries to enhance their mitigation efforts and move towards economy-wide emissions reduction targets (UNFCCC, 2015). Given this, we consider an economy-wide target to be the default unless explicitly stated otherwise. Moreover, without a clear specification of target scope, it would be unjustifiable to assume that the target covers certain sectors and GHGs but not others.

The calculation methodology for estimating emissions of GHG targets, based on target type, is described in the next section. For targets that are not economy-wide, the additional step taken to produce emissions projections for the non-covered gases and sectors are then described. Finally, the resulting economy-wide GHG emissions estimates for NDCs are outlined.

## Timeframe of GHG targets

GHG targets are usually bound to a specific implementation timeframe, either for a particular target year or for a period of time. In this technical note, "target year" refers to the year that we estimate the global GHG emissions for. While most GHG targets communicated to date have a target year of 2030 (see Appendix A), future target years are expected to be 2035, 2040, and so on.

However, some NDCs contain GHG targets for years other than the primary target year. For example, an NDC may have GHG targets for 2025, 2035, or 2050, while the primary target year for estimating global and country-level emissions is 2030. In these cases, we first estimate emissions for their respective target years, then interpolate or extrapolate the 2030 emissions level, assuming a linear trajectory between the latest historical

level and the value for the estimated target year. If an NDC provides an emissions trajectory over multiple years, including the primary target year, we use the reported values directly.

Despite the different timeframes of GHG targets, the methodology described in this technical note for estimating country-level GHG emissions should be universally applicable—regardless of target years—provided that valid data sources are available.

## Estimating country-level GHG emissions under NDCs

This section describes the methodology for estimating country-level, economy-wide GHG emissions in target years, assuming the GHG targets communicated in countries' NDCs are met in full and on time. We also discuss the method for filling the data gap for NDCs that do not communicate an economy-wide GHG target, or do not contain a GHG target at all. Given the variation in countries' NDCs, we take the following steps to estimate country-level, economy-wide GHG emissions in target years:

1. Estimate emissions based on the GHG targets as communicated in the NDCs
2. Include estimates of emissions from sectors and gases not covered in the GHG targets
3. Produce economy-wide GHG emissions estimates for all countries

### Estimating emissions based on GHG targets as communicated in NDCs

As seen above, GHG targets can take various forms, such as base-year targets, fixed-level targets, baseline scenario targets, and trajectory targets. Each type of target has particular data requirements and means of calculation.

Furthermore, if a country specifies its emissions level for a target year based on their GHG target, we use the value provided. If they do not make such a specification, we attempt to calculate the target level based on additional information provided in the NDC. For countries without GHG targets, we assume that their emissions in a target year will follow the Current Policies scenario adapted from NGFS, unless otherwise specified in the NDC, and we do not factor in the GHG impacts of non-GHG targets included in the NDCs. Those countries without GHG targets represent about 10 percent of total NDCs analyzed, and less than 3 percent of global GHG emissions (Climate Watch 2025).

Different types of GHG targets are calculated as described in Table 1.

The information provided in the NDCs for estimating target-year emissions varies. In most cases, calculating target emissions requires determining the reference level, such as historical base-year emissions or projected baseline scenario emissions. When determining reference levels, we adhere to the sector and gas coverage outlined in the GHG targets. Sectors and gases not covered by the target are assumed to follow the NGFS Current Policies scenario. If a reference level is specified in the NDC, we use the provided information. If the NDC refers to an external source document for a reference level, we extract data from that source.

Table 1 | **Calculation of target-level emissions by target type**

| TARGET TYPE | CALCULATION |
|---|---|
| **Base-year target** | ▪ Target-year emissions = Base-year emissions – (Base-year emissions × Reduction percentage) |
| **Baseline scenario target** | ▪ Target-year emissions = Projected baseline emissions – (Projected baseline emissions × Reduction percentage) |
| **Intensity target** | ▪ Target-year emissions intensity = Base-year intensity – (Base-year intensity × Reduction percentage)<br>▪ Target-year emissions = Target-year emissions intensity × Projected output in target year |
| **Fixed-level target** | ▪ Target-year emissions = Specified target emissions level |
| **Trajectory target** | ▪ Target-year emissions = Specified target emissions level along trajectory |

*Source:* Adapted from GHG Protocol (2014); Levin et al. (2015).

If the NDC does not provide the reference level (base-year or baseline scenario emissions) but indicates where to find it, we rely on official data sources including National Inventory Reports, National Communications, Biennial Update Reports, Biennial Transparency Reports, Biennial Reports, or national statistics. If the NDC references a future dataset that has not yet been published, we opt to use the latest version available. If official data sources are unavailable, we use Climate Watch historical GHG emissions data (Climate Watch 2024) for base-year emissions and NGFS Current Policies projections (Richters et al. 2024) for baseline scenario emissions. Further details on the NGFS data source are provided in the next section.

For intensity targets that require a projected output in the target year (e.g., population or GDP), if the projected output is specified in the NDC, the specified value is used directly. If not, we rely on official data sources such as Biennial Transparency Reports and national statistics. If official data is not available or is not sufficient, we use reputable resources such as the Organization for Economic Co-operation and Development's (OECD) GDP forecast (OECD 2023), and the United Nations (UN) World Population Prospects (United Nations 2024).

In some cases, the emissions level is explicitly specified in the NDC. This is particularly true for fixed-level and trajectory targets. When this occurs, the directly specified target level of emissions is used even if additional information is provided, such as reference levels that allow for calculations. In most instances, the specified target levels of emissions match the calculated values. However, if there are discrepancies, the directly specified target is used.

## Emissions from sectors and gases not covered by GHG targets

For NDCs that do not contain an economy-wide GHG target, additional calculations are needed to estimate emissions gaps from sectors and gases not covered by the target. In general, we assume that emissions from these unaccounted sectors and gases follow the Current Policies scenario based on NGFS projections (Richters et al. 2024).

The NGFS dataset is prioritized because it is one of the most up-to-date and comprehensive, compared to other data sources. There are other scenarios and sources available, however—such as the *Shared Socioeconomic Pathways Scenario Database* (SSP) and *Representative Concentration Pathways* (RCP) (Gütschow et al. 2021)—that allow researchers to explore and analyze global changes associated with emissions and the concentration of GHGs.

Box 1 | **NGFS Current Policies scenario**

The BAU scenario refers to the projected level of future emissions if current climate policies and practices remain unchanged, usually after a certain cut-off year. This scenario assumes no additional climate policies, regulations, or technological advancements aimed at reducing emissions.

NGFS Phase V's Current Policies scenario from the *Global Change Assessment Model (GCAM) 6.0* is used as the BAU to fill the emissions gaps, drawing from the approach established by Climate Resource (Meinshausen et al. 2022) for their NDCs tool. NGFS Phase V's Current Policies scenario considers existing policies reported in a variety of climate policy databases, including the Climate Policy Database (NewClimate Institute et al. 2024), as of March 2024 (NGFS 2024).

The same dataset is also used for the baseline scenario emissions level when an NDC does not provide a BAU projection for the baseline scenario target.

While the NGFS dataset does not contain country-level data for all countries, it provides emissions data for major economies and world regions, covering all seven GHGs mentioned above. The dataset also provides emissions data for most of the IPCC sectors, including energy (and energy sub-sectors such as transportation and electricity), industrial processes, and a combined agriculture and LULUCF sector (labeled Agriculture, Forestry, and Other Land Use [AFOLU]) (Richters et al. 2024).

Moreover, NGFS provides downscaled economy-wide emissions for many countries based on the GCAM 6.0 Current Policies scenario (Richters et al. 2024).[3] The downscaled dataset, however, is not as granular as the regional dataset.

The waste sector is also not covered by NGFS. We therefore use the latest country-level historical emissions data for this sector, assuming no growth rate. For AFOLU, when only one sector—either agriculture or LULUCF—is not covered by the GHG target, we apply the combined growth rate of AFOLU to fill the gap.

### Harmonization and downscaling of the NGFS Current Policies scenario

To provide granular and coherent emissions projections for all countries, we perform a simplistic harmonization and downscaling of the NGFS dataset. This includes harmonizing the two different estimations (regional and downscaled country-level data) to the common historical emissions data from Climate Watch (Climate Watch 2024), using a simple ratio method.

The NGFS regional-level output is downscaled to the national level by applying regional growth rates. The harmonization and downscaling are performed at sector and GHG level. A mapping of NGFS variables and Climate Watch data is provided in Table 2.

The downscaling is applied as follows:

The growth factor ($g$) between the target year ($t1$) and historical year ($t0$) for country $c$ is calculated for each NGFS variable—based on regional emissions data ($E$, often measured in $MtCO_2e$) available for the NGFS region ($R$) this country falls into—as follows:

$$g_c = \frac{E_{NGFS,R,t1}}{E_{NGFS,R,t0}}$$

Then the ratio is applied to Climate Watch ($CW$) historical data at $t0$ for each sector/gas, to estimate the target year ($t1$) of emissions ($E$) for country $c$:

$$E_{c,t1} = E_{CW,c,t0} \times g_c$$

A common year of 2020 ($t0$) is used for harmonization, as this is the only common historical year available across both NGFS and Climate Watch datasets. For the waste sector, where NGFS variables do not exist, we assume that the emissions level remains the same as historical Climate Watch levels in 2020.

Total GHG emissions can be obtained using either a country-level growth factor available from NGFS downscaled data, or using a regional growth factor from NGFS IAM data, subject to data availabilities. If downscaled data exist for a country, we

Table 2 | **Mapping of NGFS variables and Climate Watch data for harmonization to historical emissions**

| NGFS DATABASE | NGFS VARIABLE | CLIMATE WATCH SECTOR GAS |
|---|---|---|
| Downscaled[1] | *Emissions \| Kyoto Gases (incl. indirect LULUCF) – Emissions \| $CO_2$ \| LULUCF Indirect* | Total including LULUCF, all GHGs[3] |
| IAM[2] | Emissions \| Kyoto Gases | Total including LULUCF, all GHGs[3] |
| IAM[2] | Emissions \| $CO_2$ \| Energy | Energy $CO_2$ |
| | Emissions \| $CH_4$ \| Energy | Energy $CH_4$ |
| | Emissions \| $N_2O$ \| Energy | Energy $N_2O$ |
| | Emissions \| Kyoto Gases \| Industry | Industrial Processes $CO_2$ |
| | Emissions \| Kyoto Gases \| Industry | Industrial Processes $CH_4$ |
| | Emissions \| Kyoto Gases \| Industry | Industrial Processes $N_2O$ |
| | Emissions \| F-Gases | Industrial Processes F-Gas |
| | Emissions \| $CH_4$ \| AFOLU | Agriculture $CH_4$ |
| | Emissions \| $N_2O$ \| AFOLU | Agriculture $N_2O$ |
| | Emissions \| $CO_2$ \| AFOLU | Land-Use Change and Forestry $CO_2$ |
| | Emissions \| $CH_4$ \| AFOLU | Land-Use Change and Forestry $CH_4$ |
| | Emissions \| $N_2O$ \| AFOLU | Land-Use Change and Forestry $N_2O$ |
| n/a | n/a | Waste $CH_4$ |
| n/a | n/a | Waste $N_2O$ |

*Notes:* 1. The NGFS Downscaled dataset provides country-level emissions data for over 190 countries. The dataset is only used for calculating total GHG emissions (Kyoto Total). 2. The NGFS IAM dataset refers to the main database used to obtain modeled data. It provides data for a few countries and regions. The full list of country and regional data availability, and the mapping of NGFS countries and regions are provided in Appendix B. 3. When Kyoto Total (total GHG emissions including LULUCF, covering all GHGs) can be calculated from either the NGFS IAM or Downscaled datasets, we prioritize calculations based on the NGFS Downscaled dataset. This is because it provides a better proxy for country-level changes than applying regional growth factors based on the NGFS IAM dataset.

*Source:* Authors. Compiled from Richters et al. (2024) and Climate Watch (2024).

prioritize the total GHG emissions calculated from NGFS downscaled data, as the country-level growth factor would be a better approximation to country-level changes compared to a regional growth factor.

## Economy-wide GHG emissions

After we estimate the GHG emissions as communicated in NDC targets, and obtain emissions projections data for non-covered sectors and GHGs, we are able to calculate the economy-wide net GHG emissions in a target year for all countries (Table 3).

For NDCs with an economy-wide GHG target (Type 1), the target levels of GHG emissions estimates are used without any further calculations.

For NDCs with GHG targets that only cover partial sectors or particular gases (Type 2), we use the NGFS Current Policies trajectory (downscaled from regional values if needed) for non-covered sectors and gases. This allows us to complete the GHG target as communicated in their NDCs.

For the NDCs that do not contain a GHG emission target (Type 3), we assume that the country will follow the economy-wide Current Policies trajectory downscaled and/or harmonized based on the NGFS emissions trajectory.

## Additional assumptions and special treatments

NDCs often present targets in complex ways. Some NDCs include targets with varied conditions, while others may provide targets with ranges, or contain multiple targets. We thus apply additional assumptions to standardize the approach taken for such cases.

### Unconditional and conditional targets

NDCs may contain GHG targets that are unconditional, conditional, or both. Unconditional targets are not dependent on external support, whereas conditional targets require international assistance such as financial support and capacity building. Both types of GHG targets are estimated using the same methodologies discussed earlier.

For NDCs that include only unconditional targets, these are used for both unconditional and conditional estimates, as they represent the highest level of GHG emissions the country is committed to achieve. For NDCs that only specify conditional targets, we assume a Current Policies emissions trajectory in place of an unconditional target, since it reflects the possible emissions trajectory without further mitigation measures.

### NDCs with targets expressed as a range

For GHG targets expressed as a range, the methodology described above can be used to estimate emissions for both ends of the range. The mid-point of the target range is used for both country-level reporting and global aggregation.

### NDCs with multiple types of GHG targets

NDCs can have multiple GHG targets. For example, an NDC can have both a base-year target and an emissions budget target, or an intensity target and peaking target. In those cases, we attempt to estimate the emissions level based on each target type, with priority given to the GHG target that provides the most detailed information; that is, the one that requires the least assumptions for its calculation. When the target level is different across target types, we assume the most ambitious target as the target level of emissions, under the assumption that other targets will contribute to the achievement of the most ambitious one (Fransen et al. 2021).

Table 3 | **NDC categories by target and calculation method**

| NDC CATEGORY | DESCRIPTION | GHG EMISSIONS CALCULATION METHODOLOGY |
|---|---|---|
| Type 1 | NDCs with economy-wide GHG target | GHG emissions in target year, estimated from the GHG target as communicated in the NDCs |
| Type 2 | NDCs with GHG target but not economy-wide | GHG emissions in target year, estimated from the GHG target as communicated in the NDCs + the NGFS Current Policies trajectory (downscaled/harmonized) for missing sectors and/or gases |
| Type 3 | NDCs with no GHG target | NGFS Current Policies trajectory for all sectors and gases (downscaled/harmonized) |

*Source:* Authors.

### NDCs with a single GHG target framed in different ways

Some NDCs may provide information that allows for calculations to be framed in more than one way (for example, a target framed in both percentage reduction and absolute emissions reduction). In most cases, the numbers from different calculations align closely with each other, with minor mathematical differences most likely due to rounding. When this happens, we prioritize the target framing which requires fewer calculations. Specifically, absolute emissions values are preferred over percentage reductions, and directly stated target levels are prioritized over calculated values.

### NDCs with internally contradictory information

In rare cases, NDCs contain internally inconsistent information. For example, an NDC might present target level emissions in a chart as one value, while calculations based on information provided in a table yield a different value. When this occurs, we exercise caution and treat the NDCs on a case-by-case basis to obtain the mostly like accurate presentation of the target. At the same time, we follow the rule of prioritizing the most straightforward calculation possible.

### Countries without NDCs

Countries without NDCs are treated the same as NDCs with no GHG targets (Type 3) and are assumed to follow an NGFS Current Policies trajectory. As of the writing of this technical note, Iran and Yemen have submitted Intended Nationally Determined Contributions (INDCs), but not NDCs. Libya has not submitted any document. For these three countries, the downscaled Current Policies trajectory from NGFS is applied.

## Estimating global GHG emissions under NDCs

The global GHG emissions level in a target year is determined by aggregating the country-level GHG emissions calculated above with emissions that are not covered by countries' NDCs.

For emissions that are not covered by a specific country—namely emissions from international shipping and aviation bunker fuels—we assume the achievement of climate commitments by respective international organizations (UNFCCC 2024a):

- **Emissions from international shipping:** emissions for international shipping at 2030 are estimated as meeting the targets set in the revised IMO *GHG Strategy* (IMO 2024). The indicative checkpoint is to reduce the total annual GHG emissions from international shipping by at least 20 percent—striving for 30 percent—by 2030, and by at least 70 percent—striving for 80 percent—by 2040, compared to 2008 levels. The less ambitious bound of the indicative target (20 percent) for 2030 was used.
- **Emissions from international aviation:** emissions for international aviation assume implementation of ICAO's Carbon Offsetting and Reduction Scheme for International Aviation (CORSIA) (ICAO 2024). The target adopted is a 5 percent reduction by 2030 through the use of sustainable aviation fuels. Reference emissions for the CORSIA baseline (2024–2035) are set at 85 percent of 2019 levels.

The list of countries with emissions inventories available does not contain all territories and regions of the world that could be sources of GHG emissions. To address those missing GHG emissions, we apply the difference between reported world total GHG emissions and the sum of country-level GHG emissions, drawing from Climate Watch data.

- **Emissions from territories not covered by country-level inventories:** This analysis covers only parties to the UNFCCC, leaving out some territories whose anthropogenic emissions might be included in world total GHG emissions. The difference is estimated at around 0.45 $GtCO_2e$, based on average Climate Watch historical data over the last 10 years.

Lastly, we adjust for the differences in LULUCF emissions between national inventories and bookkeeping models to ensure consistency with results from integrated assessment models, following an existing approach (Kuramochi et al. 2024; Grassi et al. 2022). Recent research (Grassi et al. 2023) confirms that there is a significant discrepancy of 6.7 $GtCO_2$ per year in global land use $CO_2$ emissions between the bookkeeping models (4.8 $GtCO_2$) and national greenhouse gas inventories (NGGI) (sink of –1.9 $GtCO_2$) for the period 2000–2010. To account for the difference, we apply the 6.7 $GtCO_2$ to the total global GHG emissions.

## Limitations

This technical note presents a methodology for estimating global and country-level GHG emissions based on the targets outlined in countries' NDCs. Such an estimate is a pre-requisite to assessing target ambition and implementation, though the methodology itself does not extend to those assessments (Fransen et al. 2023). It focuses on the effects of achieving GHG targets, assuming that non-GHG targets and other NDC measures contribute to achieving GHG targets.

Data gaps in NDCs impose limitations on the estimates generated by this methodology at both country and global levels. At the country level, key sources of uncertainty include assumptions about the scope and coverage of the targets when not explicitly stated in the NDCs. Furthermore, the modeled emissions projections used to fill NDC data gaps are subject to their own uncertainties, based on respective underlying methodologies.

Affecting both our country-specific and global estimates is our treatment of Global Warming Potential (GWP) values, which are used to convert emissions of non-$CO_2$ gases to $CO_2$-equivalent emissions. IPCC reports have produced multiple GWP values for each non-$CO_2$ gas (Ramaswamy et al. 1996; Forster et al. 2007; Myhre et al. 2013; Forster et al. 2021). Ideally, data from NDCs and the data sources used to fill country- and global-level data gaps would be converted into a common GWP. Several factors limit the feasibility of such harmonization. First, more than half of currently submitted NDCs[4] do not specify which GWP they are based on. Second, even for NDCs that do specify GWP values, converting the values would require making an assumption about the share of total target-year emissions pertaining to each gas, as this breakdown is typically not available in the NDC itself. A summary of GWP values specified in the NDCs are provided in Appendix A. In our view, these limitations—combined with the fact that they do not affect $CO_2$, the predominant gas affected by NDCs—obviate the utility of attempting to convert to a consistent GWP value.

As countries continue to communicate updated NDCs and introduce new GHG targets, we will continue to examine these for opportunities to enhance the accuracy and comprehensiveness of future emissions estimates. Furthermore, under the Paris Agreement's enhanced transparency framework (UNFCCC 2025), countries are required to submit Biennial Transparency Reports (BTRs). These BTRs will contain valuable information for tracking the progress of the NDCs every two years. We anticipate incorporating BTR information into our analysis in future iterations. As such, this methodology should be viewed as dynamic and subject to revisions and improvements as new data and scientific insights become available. The Climate Watch online data platform will be updated frequently to represent the latest GHG targets and information available.

# Appendices

## Appendix A: Summary of NDCs analyzed as of November 2024

Analysis has been conducted for the latest NDC submission for each country, for a total of 171 NDCs. As of November 2024, 195 out of the 198 parties of the UN Framework Convention on Climate Change (UNFCCC) have submitted an NDC. The European Union and its 27 member states submitted the same NDC document and thus are considered as one. Iran and Yemen have submitted Intended Nationally Determined Contributions (INDCs) only, but not NDCs. Libya has not submitted any document.

Among the 168 NDCs, 70 are considered economy-wide (representing 48 percent of global emissions) and did not need gap filling. Eighty-two (representing 40 percent of global emissions) have 2030 targets that do not cover all sectors and gases, requiring additional calculations to address their emission gap. Finally, 16 do not contain a 2030 GHG target, requiring the use of Current Policies emissions projections as a substitute.

Table A-1 summarizes the types of NDCs and consistencies of GWP values under each type. At the country level, emissions values for Type 1 countries are internally consistent, emissions values for Type 2 countries may be based on internally inconsistent GWP values, and emissions values for Type 3 countries are all based on AR4 values. Global values thus reflect a mix of GWP values.

Following the methodology discussed in this technical note, the total global GHG emissions in 2030 comes to 55.8 $GtCO_2e$ under unconditional NDCs, and 52.6 $GtCO_2e$ under conditional NDCs. The total includes:

- Emissions under sectors and GHGs covered by NDCs: 36.6 $GtCO_2e$ unconditional, 38.4 $GtCO_2e$ conditional
- Emissions from sectors and GHGs not covered by NDCs: 11.1 $GtCO_2e$ unconditional, 6.0 $GtCO_2e$ conditional
- Emissions from international bunkers: 1.0 $GtCO_2e$
- Emissions from territories not covered by countries that are party to the UNFCCC: 0.4 $GtCO_2e$
- Adjustment of LULUCF: 6.7 $GtCO_2e$

Table A-1 | **NDC categories and GWP under different NDC categories**

| NDC CATEGORY | DESCRIPTION | GWP CONSISTENCY | SUMMARY OF CATEGORY |
|---|---|---|---|
| Type 1 | NDCs with economy-wide GHG target | Estimates for individual NDCs in this category are expressed in whichever GWP value the reference value uses. Most reference values are sourced from the NDCs directly. Provided the reference values use an internally consistent GWP value, these values are internally consistent within each country but vary across countries. | 70 NDCs, among which:<br>▪ 11 specified AR4 GWP values<br>▪ 39 specified non-AR4 GWP values (AR2, AR3, and AR5)<br>▪ 20 with unspecified GWP values |
| Type 2 | NDCs with GHG target but not economy-wide | Estimates for individual NDCs in this category include data from Climate Watch and/or NGFS—expressed in AR4—as well as data from the individual NDC, which are expressed in whichever GWP value the NDC uses. This may result in internal inconsistency in cases where the NDC does not use AR4. | 82 NDCs, among which:<br>▪ 15 specified AR4 GWP values<br>▪ 27 with non-AR4 GWP values (AR2 and AR5), representing 270 $MtCO_2e$ emissions filled for a total aggregated 3.6 $GtCO_2e$ GHG emissions in 2030<br>▪ 40 with unspecified GWP values, representing 3.2 $GtCO_2e$ of emissions filled towards a total aggregated 18.9 $GtCO_2e$ emissions |
| Type 3 | NDCs with no GHG target | Estimates for this category are sourced entirely from Climate Watch and/or NGFS, thus are expressed in AR4 and are internally consistent within each country and across countries within this category. | 16 NDCs, among which:<br>▪ 2 specified non-AR4 GWP values (AR2), and 14 with unspecified GWP values<br>▪ 3 countries without an NDC |

*Source:* Author calculations, drawing from Climate Watch (2025) data.

## Appendix B: NGFS Regions and Economies

The following table outlines regional groups reported under the NGFS Current Policies scenario and countries that fall under each region.

Table B-1 | **List of countries mapped to NGFS countries and regions**

| COUNTRY | ISO | NGFS DOWNSCALED DATA AVAILABILITY | NGFS IAM REGION |
|---|---|---|---|
| Afghanistan | AFG | | South Asia |
| Albania | ALB | ALB | Europe_Non_EU |
| Algeria | DZA | DZA | Africa_Northern |
| Andorra | AND | | EU-15 |
| Angola | AGO | AGO | Africa_Southern |
| Antigua and Barbuda | ATG | | Central America and Caribbean |
| Argentina | ARG | ARG | Argentina |
| Armenia | ARM | ARM | Central Asia |
| Australia | AUS | AUS | Australia_NZ |
| Azerbaijan | AZE | AZE | Central Asia |
| Bahrain | BHR | BHR | Middle East |
| Bangladesh | BGD | BGD | South Asia |
| Barbados | BRB | | Central America and Caribbean |
| Belarus | BLR | BLR | Europe_Eastern |
| Belize | BLZ | | Central America and Caribbean |
| Benin | BEN | BEN | Africa_Western |
| Bhutan | BTN | | South Asia |
| Bolivia | BOL | BOL | South America_Southern |
| Bosnia and Herzegovina | BIH | BIH | Europe_Non_EU |
| Botswana | BWA | BWA | Africa_Southern |
| Brazil | BRA | BRA | Brazil |
| Brunei | BRN | BRN | Southeast Asia |
| Burkina Faso | BFA | | Africa_Western |
| Burundi | BDI | | Africa_Eastern |
| Cambodia | KHM | KHM | Southeast Asia |
| Cameroon | CMR | CMR | Africa_Western |
| Canada | CAN | CAN | Canada |
| Cape Verde | CPV | | Africa_Western |
| Central African Republic | CAF | | Africa_Western |

Table B-1 | **List of countries mapped to NGFS countries and regions (cont.)**

| COUNTRY | ISO | NGFS DOWNSCALED DATA AVAILABILITY | NGFS IAM REGION |
|---|---|---|---|
| Chad | TCD | | Africa_Western |
| Chile | CHL | CHL | South America_Southern |
| China | CHN | CHN | China |
| Colombia | COL | COL | Colombia |
| Comoros | COM | | Africa_Eastern |
| Cook Islands | COK | | Southeast Asia |
| Costa Rica | CRI | CRI | Central America and Caribbean |
| Cuba | CUB | CUB | Central America and Caribbean |
| Democratic Republic of the Congo | COD | COD | Africa_Western |
| Djibouti | DJI | | Africa_Eastern |
| Dominica | DMA | | Central America and Caribbean |
| Dominican Republic | DOM | DOM | Central America and Caribbean |
| Ecuador | ECU | ECU | South America_Southern |
| Egypt | EGY | EGY | Africa_Northern |
| El Salvador | SLV | SLV | Central America and Caribbean |
| Equatorial Guinea | GNQ | | Africa_Western |
| Eritrea | ERI | ERI | Africa_Eastern |
| Eswatini | SWZ | | Africa_Southern |
| Ethiopia | ETH | ETH | Africa_Eastern |
| European Union (27) | EUU | | EU-12, EU-15 |
| Federated States of Micronesia | FSM | | Southeast Asia |
| Fiji | FJI | | Southeast Asia |
| Gabon | GAB | GAB | Africa_Western |
| Gambia | GMB | | Africa_Western |
| Georgia | GEO | GEO | Central Asia |
| Ghana | GHA | GHA | Africa_Western |
| Grenada | GRD | | Central America and Caribbean |
| Guatemala | GTM | GTM | Central America and Caribbean |
| Guinea | GIN | | Africa_Western |
| Guinea Bissau | GNB | | Africa_Western |
| Guyana | GUY | | South America_Northern |
| Haiti | HTI | HTI | Central America and Caribbean |

Table B-1 | **List of countries mapped to NGFS countries and regions (cont.)**

| COUNTRY | ISO | NGFS DOWNSCALED DATA AVAILABILITY | NGFS IAM REGION |
|---|---|---|---|
| Holy See (Vatican City State) | VAT | | EU-15 |
| Honduras | HND | HND | Central America and Caribbean |
| Iceland | ISL | ISL | European Free Trade Association |
| India | IND | IND | India |
| Indonesia | IDN | IDN | Indonesia |
| Iran | IRN | IRN | Middle East |
| Iraq | IRQ | IRQ | Middle East |
| Israel | ISR | ISR | Middle East |
| Ivory Coast | CIV | CIV | Africa_Western |
| Jamaica | JAM | JAM | Central America and Caribbean |
| Japan | JPN | JPN | Japan |
| Jordan | JOR | JOR | Middle East |
| Kazakhstan | KAZ | KAZ | Central Asia |
| Kenya | KEN | KEN | Africa_Eastern |
| Kiribati | KIR | | Southeast Asia |
| Kuwait | KWT | KWT | Middle East |
| Kyrgyzstan | KGZ | KGZ | Central Asia |
| Laos | LAO | | Southeast Asia |
| Lebanon | LBN | LBN | Middle East |
| Lesotho | LSO | | Africa_Southern |
| Liberia | LBR | | Africa_Western |
| Libya | LBY | LBY | Africa_Northern |
| Liechtenstein | LIE | | European Free Trade Association |
| Madagascar | MDG | | Africa_Eastern |
| Malawi | MWI | | Africa_Southern |
| Malaysia | MYS | MYS | Southeast Asia |
| Maldives | MDV | | South Asia |
| Mali | MLI | MLI | Africa_Western |
| Marshall Islands | MHL | | Southeast Asia |
| Mauritania | MRT | | Africa_Western |
| Mauritius | MUS | MUS | Africa_Eastern |
| Mexico | MEX | MEX | Mexico |

Table B-1 | **List of countries mapped to NGFS countries and regions (cont.)**

| COUNTRY | ISO | NGFS DOWNSCALED DATA AVAILABILITY | NGFS IAM REGION |
|---|---|---|---|
| Moldova | MDA | MDA | Europe_Eastern |
| Monaco | MCO | | EU-15 |
| Mongolia | MNG | MNG | Central Asia |
| Montenegro | MNE | MNE | Europe_Non_EU |
| Morocco | MAR | MAR | Africa_Northern |
| Mozambique | MOZ | MOZ | Africa_Southern |
| Myanmar | MMR | MMR | Southeast Asia |
| Namibia | NAM | NAM | Africa_Southern |
| Nauru | NRU | | Southeast Asia |
| Nepal | NPL | NPL | South Asia |
| New Zealand | NZL | NZL | Australia_NZ |
| Nicaragua | NIC | NIC | Central America and Caribbean |
| Niger | NER | NER | Africa_Western |
| Nigeria | NGA | NGA | Africa_Western |
| Niue | NIU | | Southeast Asia |
| North Korea | PRK | PRK | Southeast Asia |
| North Macedonia | MKD | MKD | Europe_Non_EU |
| Norway | NOR | NOR | European Free Trade Association |
| Oman | OMN | OMN | Middle East |
| Pakistan | PAK | PAK | Pakistan |
| Palau | PLW | | Southeast Asia |
| Palestine | PSE | | Middle East |
| Panama | PAN | PAN | Central America and Caribbean |
| Papua New Guinea | PNG | | Southeast Asia |
| Paraguay | PRY | PRY | South America_Southern |
| Peru | PER | PER | South America_Southern |
| Philippines | PHL | PHL | Southeast Asia |
| Qatar | QAT | QAT | Middle East |
| Republic of Congo | COG | COG | Africa_Western |
| Republic of Serbia | SRB | SRB | Europe_Non_EU |
| Russia | RUS | RUS | Russia |
| Rwanda | RWA | | Africa_Eastern |

Table B-1 | **List of countries mapped to NGFS countries and regions (cont.)**

| COUNTRY | ISO | NGFS DOWNSCALED DATA AVAILABILITY | NGFS IAM REGION |
|---|---|---|---|
| Saint Kitts and Nevis | KNA | | Central America and Caribbean |
| Saint Lucia | LCA | | Central America and Caribbean |
| Saint Vincent and the Grenadines | VCT | | Central America and Caribbean |
| Samoa | WSM | | Southeast Asia |
| San Marino | SMR | | EU-15 |
| São Tomé and Príncipe | STP | | Africa_Western |
| Saudi Arabia | SAU | SAU | Middle East |
| Senegal | SEN | SEN | Africa_Western |
| Seychelles | SYC | | Southeast Asia |
| Sierra Leone | SLE | | Africa_Western |
| Singapore | SGP | SGP | Southeast Asia |
| Solomon Islands | SLB | | Southeast Asia |
| Somalia | SOM | | Africa_Eastern |
| South Africa | ZAF | ZAF | South Africa |
| South Korea | KOR | KOR | South Korea |
| South Sudan | SSD | SSD | Africa_Eastern |
| Sri Lanka | LKA | LKA | South Asia |
| Sudan | SDN | SDN | Africa_Eastern |
| Suriname | SUR | SUR | South America_Northern |
| Switzerland | CHE | CHE | European Free Trade Association |
| Syria | SYR | SYR | Middle East |
| Tajikistan | TJK | TJK | Central Asia |
| Thailand | THA | THA | Southeast Asia |
| The Bahamas | BHS | | Central America and Caribbean |
| Timor-Leste | TLS | | Southeast Asia |
| Togo | TGO | TGO | Africa_Western |
| Tonga | TON | | Southeast Asia |
| Trinidad and Tobago | TTO | TTO | Central America and Caribbean |
| Tunisia | TUN | TUN | Africa_Northern |
| Türkiye | TUR | TUR | Europe_Non_EU |
| Turkmenistan | TKM | TKM | Central Asia |
| Tuvalu | TUV | | Southeast Asia |

Table B-1 | **List of countries mapped to NGFS countries and regions (cont.)**

| COUNTRY | ISO | NGFS DOWNSCALED DATA AVAILABILITY | NGFS IAM REGION |
|---|---|---|---|
| Uganda | UGA | UGA | Africa_Eastern |
| Ukraine | UKR | UKR | Europe_Eastern |
| United Arab Emirates | ARE | ARE | Middle East |
| United Kingdom | GBR | GBR | EU-15 |
| United Republic of Tanzania | TZA | TZA | Africa_Southern |
| United States of America | USA | USA | USA |
| Uruguay | URY | URY | South America_Southern |
| Uzbekistan | UZB | UZB | Central Asia |
| Vanuatu | VUT | | Southeast Asia |
| Venezuela | VEN | VEN | South America_Northern |
| Vietnam | VNM | VNM | Southeast Asia |
| Yemen | YEM | YEM | Middle East |
| Zambia | ZMB | ZMB | Africa_Southern |
| Zimbabwe | ZWE | ZWE | Africa_Southern |

*Source:* Richters et al. 2024.

## Endnotes

1. Median value across modeling scenarios limiting global temperature increase to below 1.5°C with no or limited overshoot (>50 percent chance) by 2100.
2. The percentages from the *Emissions Gap Report* 2024 (UNEP 2024) differ slightly from the reduction needed (43 percent) stated in the IPCC *Sixth Assessment Report* (AR6) (IPCC 2022), as the UNEP *Emissions Gap Report* uses updated 2019 emissions levels compared with the AR6.
3. Referred to as *Downscaling NGFS GCAM 6.0*. The list of countries for which NGFS provides downscaled economy-wide emissions is in Appendix B.
4. Seventy-three NDCs as of November 2024, covering approximately half of global emissions, did not specify GWP value.

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